

8. Holography has been used to make archival recordings of valuable and/or fragile museum artifacts.

9. Sony Electronics uses holographic technology in their digital cameras. A holographic crystal allows the camera to detect the edge of the subject and differentiate between it and the background. As a result, the camera is able to focus accurately in dark conditions.

10. Holography has been used by artists to create pulsed holographic portraits as well as other works of art.

11. Future colour LCDs will be brighter and whiter as a result of holographic technology.

12. Holographic night-vision goggles

13. Many researchers believe that holographic televisions will become available within a decade. Holographic motion pictures have been previously attempted and may become a reality within the next few years.

14. Holographic memory is a new optical storage method that can store one terabyte (1000GB) of data in a crystal approximately the size of a sugar cube. In comparison, CDs can hold 650-700MB, DVDs 4.7GB, and computer hard drives 120GB.

15. Optical computers will be capable of delivering trillions of bits of information faster than the latest computers.

Challenges and solutions

Holograms don't move. To make them move, we need to create a hologram in real time. This needs little tiny pixels—smaller than anything we can build in large volume at low cost. Although the holographic displays research community has made significant advances in recent years, the prospect of a true holographic display still remains elusive. Holographic displays are still at the basic research stage and require significant technological advances before they become commercially feasible.

People expect that high-quality holograms will entertain them in the near future, because visualisations are increasingly becoming sophisticated and highly imaginative due to the use of computer-aided graphics and recently-developed digital devices



3D LED light boxes (Credit: www.w-co.co.uk)

es that provide augmented or virtual reality. To be commercially feasible in a range of applications—from medicine to gaming to media—the hologram challenge is daunting. It involves scaling an electronic device to a size small enough to fit on a table top, while making it robust enough to render immense amounts of data needed to create a full-surround 3D viewing experience from every angle—without the need for special glasses or other viewing aids.

The capacity of holographic-video displays is measured according to the space-bandwidth (SB) product. The SB product is the product of the number of pixels and their spatial frequency—the inverse of their size. So we are looking for a large number of very small pixels. To have a large number of pixels is not really helpful if these are large.

One of the big barriers to the consumer application of holographic technology has been the price of the components. Affordable holographic displays could easily be valuable to the gaming and entertainment industries, and could add a new dimension to videoconferencing.

Some of the limitations and their possible solutions for electronics holographic displays are summarised below.

Image quality. Digital holograms are often associated with noisy, low-contrast, low-resolution and mono-colour imagery. High-contrast, high-resolution and full-colour digital

holograms can be formed using existing hardware devices, yielding images that more closely match the quality of conventional displays.

Computation speed. Due to the complexity of required calculations, digital holograms are also often associated with slow, off-line calculations. However, eye-tracked approximate holograms provide correct image focus and best image quality.

Focus control. The ability to change the image focus is desirable in a near-eye display: It addresses the accommodation-convergence conflict, allows image focus to match eye focus in a see-through display, and adds realism. Unlike vari-focal, multi-focal and light field displays, a holographic display is able to provide per-pixel focus control with virtually no discretisation, enabling smooth and natural focal cues to imagery.

Vision correction. Another desirable quality of a display is vision correction—the ability to fix defects in the user's vision. Such a display allows users to view the display without their glasses. Recent holographic displays can correct simple vision problems such as near- and far-sightedness as well as higher-order vision problems like astigmatism.

Aberration correction. Holographic displays are also capable of aberration correction—encoding optical corrections to the display